Compactness and contradiction

Terence Tao · 1 July 2011

https://terrytao.wordpress.com/2011/07/01/compactness-and-contradiction/

tao2typ · converted to Typst 2026-10-14 (UTC)

I recently finished the first draft of the last of my books based on my 2010 blog posts (and also my Google buzzes), entitled "Compactness and contradiction". The PDF of this draft is available here. This is a somewhat assorted (and lightly edited) collection of posts (and buzzes), though concentrating in the areas of analysis (both standard and nonstandard), logic, and group theory. As always, comments and corrections are welcome.

Comments (10)



Bo Jacoby · 1 July 2011

First of all: Thank you for giving me this book. I look forward to studying it.
I looked at the Contents, and I would like to tell you about (my) Ordinal Fractions. They are effective for numbering chapters and paragraphs and anything else. They are too elementary to catch the attention of mathematicians.
The digit zero has two meanings. In '1770' zero means 'nothing', and in 'the 1770s' zero means 'everything'.
In order to avoid misunderstandings, zero should mean only one thing, and in an ordinal fraction, zero means 'everything'.
There are only 9 one digit numbers left when zero doesn't count. So this is not decimal, it is most nonal.
There is no dot between the chapter number and the section number. When there are more than 9 sections in a chapter the sections are numbered with two (nonzero) digits.
An ordinal fraction is something like 'the third fourth', because 'the third' is an ordinal number and 'a fourth' is a fraction. Like this:
00 whole
01 odd fourths
02 even fourths
10 first half
11 first fourth
12 second fourth
20 second half
21 first fourth
22 second fourth
The five ordinal fraction relations are these:
1 The first half is equal to the first half: 10=10
2 The first half is part of the whole: 1011
4 The first half is parallel with the second half: 10><20
5 The first half intersects the odd fourths: 1001
Ordinal fractions are and'ed like this:
10+10=10
10+00=10
10+11=11
10+20=Ø

10+01=11
10+Ø=Ø
Ø is the empty set or the impossible condition.
Ø is the improper ordinal fraction.
So your table of contents may look like below,
Note that 2122<2100, meaning that the section on circular arguments are part of chapter on logic and foundations.
Yours truly, Bo.
0000 Compactness and contradiction
1100 Preface
1200 A remark on notation
1300 Acknowledgments
2100 Logic and foundations
2111 Material implication
2112 Errors in mathematical proofs
2113 Mathematical strength
2114 Stable implications
2115 Notational conventions
2121 Abstraction
2122 Circular arguments
2123 The classical number systems
2124 Round numbers
2125 The \no self-defeating object" argument, revisited
2131 The \no self-defeating object" argument, and the vagueness paradox
2132 A computational perspective on set theory
2200 Group theory
2211 Torsors
2212 Active and passive transformations
2213 Cayley graphs and the geometry of groups
2214 Group extensions
2215 A proof of Gromov's theorem
2300 Analysis
2311 Orders of magnitude, and tropical geometry
2312 Descriptive set theory vs. Lebesgue set theory
2313 Complex analysis vs. real analysis
2314 Sharp inequalities
2315 Implied constants and asymptotic notation
2321 Brownian snowflakes
2322 The Euler-Maclaurin formula, Bernoulli numbers, the zeta function, and real-variable analytic continuation
2323 Finitary consequences of the invariant subspace problem
2324 The Guth-Katz result on the Erd}os distance problem
2325 The Bourgain-Guth method for proving restriction theorems
2400 Nonstandard analysis
2411 Real numbers, nonstandard real numbers, and nite precision arithmetic
2412 Nonstandard analysis as algebraic analysis
2413 Compactness and contradiction: the correspondence principle in ergodic theory
2414 Nonstandard analysis as a completion of standard analysis
2425 Concentration compactness via nonstandard analysis

2500 Partial di erential equations
2511 Quasilinear well-posedness
2512 A type diagram for function spaces
2513 Amplitude-frequency dynamics for semilinear dispersive equations
2514 The Euler-Arnold equation
2600 Miscellaneous
2611 Multiplicity of perspective
2612 Memorisation vs. derivation
2613 Coordinates
2614 Spatial scales
2615 Averaging
2621 What colour is the sun?
2622 Zeno's paradoxes and induction
2623 Jevons' paradox
2624 Bayesian probability
2625 Best, worst, and average-case analysis
2631 Duality
2632 Open and closed conditions
3100 Bibliography
3200 Index

Matthew N. Petersen · 1 July 2011

Pg. 18, you have “Gdel’s incompleteness theorem”, rather than “Gödel’s…”.

Matthew N. Petersen · 1 July 2011

Same difficulty on p. 29
[Corrected, thanks – T.]



gowers · 1 July 2011

A quick typo: in the statement of Theorem 4.3.11 you write “Szemerdi”. An easy exercise in reverse engineering suggests, in the light of the “Gdel” typo mentioned above, that something more general has gone wrong …
[Ah, some of the text I incorporated to the book contained accents which LaTeX then refused to recognise. I think the problem is fixed now and will be corrected in the next revision of the ms – T.]



Greg · 4 July 2011

Couple of typos right at the start. On page 2, in statement (8) you have “Is A, then B” which looks like it should be “If A, then B”. Lower down, penultimate paragraph (just before the footnote 1) your implication arrows seem to have gone wrong (-> rather than -> for A \implies B and B \implies A)
Otherwise, this looks good so far!
[Thanks, this will be corrected in the next version of the ms. -T.]



Greg · 4 July 2011

One more before bed: on page 9, top paragraph, you have “the two are not symmetric, because in both cases is bounding an unknown quantity by a known quantity.” The “is” in the second clause is missing its subject — we’re not told _what_ is bounding the unknown quantity.
–g
[Thanks, this will be corrected in the next version of the ms. -T.]

John · 4 July 2011

Page 80: “algberaic” -> “algebraic”

[Thanks, this will be corrected in the next version of the ms. -T.]

diego.maldona@gmail.com · 5 July 2011

Dear Terry, thank you for posting the book. It’s always insightful to read your unifying take on so many, seemingly disconnected, Math topics. Below are a few typos I found.

p.13, line 8, misplaced comma

p.20, line 16, “doesnt”

p.22, line -13, “and its proof is” should be “and its proof are”

p.22, last line, extra semicolon

p.23, “If B was an element” should read “If B were an element”

p.29. The sentence starting with “However...” goes on for ten and a half lines and there is a “then” that’s not quite working out.

p.29, first footnote, I believe “boolean” should be “Boolean” (as “Bayesian” reads “Bayesian”). “boolean” also appears in other parts of the manuscript. Same issue with “gaussian”; but, apparently, “abelian” is ok.

p.30, line 8, “that A was finite” should be “that A were finite” or “that A is finite”

p.48, line 8, “translation” should be “translations”

p.110, There are several “Erdos” and “Holder” (without the umlaut).

p.110, line 11, “considing”

p.192, lines -5, -4, the parameter “p” should be in math font

In Section 1.11, after discussing Richard’s paradox (p.32), you could mention that if a reader runs into a genie willing to grant three wishes, the instruction “I wish for 1000 wishes” won’t have any effect, since that’s not a wish, but a meta-wish. This situation could also be phrased as a paradox (writing, for instance, that the genie will grant exactly 3 wishes).

Regarding paradoxes, it will be illuminating to have your take on the “surprise test paradox”

http://en.wikipedia.org/wiki/Unexpected_hanging_paradox

http://arxiv.org/abs/math/9903160

I loved the “digital-images analogy” when illustrating the concepts of extension and quotient.

[Thanks for the corrections! They will be incorporated in the next revision of the ms. The surprise test paradox is discussed on this blog at https://terrytao.wordpress.com/2011/05/19/epistemic-logic-temporal-epistemic-logic-and-the-blue-eyed-islander-puzzle-lower-bound/ -T.]



Hemant Verma · 9 July 2011

Hi All,

There are related mathematics and algorithms problem here, for those who love mathematics / algorithm problem solviing Mathalon , after all whats in mathematics without problems.

-Hemant

Anonymous · 26 September 2016

Dear Terry!

What’s the point of thinking of an implication A –> B as asserting that B is at least as true as A in order to understand the disjunction elimination? In know that “A or B” has the biggest truth value of the two truth values of A and B. Thus if C is at least as true as A and also at least as true as B, then “A or B” is at least as true as C (since “A or B” has the same truth value as A or as B). But why does this shed light on the disjunction elimination?